2.30 Integration: The Trapezium Rule

Worked Solutions 1–23

1)

A will produce an overestimate as the curve is convex (concave up). The top of each trapezium will be above the curve.

B will produce an underestimate as the curve is concave (concave down). The top of each trapezium will be below the curve.

C will produce the exact result as the function is linear.

D will produce an underestimate as the curve is concave (concave down). The top of each trapezium will be below the curve.

2)

9 ordinates implies 8 strips. So $h = \frac{8-3}{8} = \frac{5}{8} = 0.625$

3 a)

$h = \frac{3-0}{3} = 1$

x	0	1	2	3
y	0	1	4	9

$A = \frac{1}{2} \times 1 \times (0 + 9 + 2(1 + 4)) = \frac{19}{2}$

3 b)

$h = \frac{4-0}{4} = 1$

x	0	1	2	3	4
y	1	2	5	10	17

$A = \frac{1}{2} \times 1 \times (1 + 17 + 2(2 + 5 + 10)) = 26$

3 c)

$h = \frac{2-0}{4} = \frac{1}{2}$

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
y	0	$\frac{1}{8}$	1	$\frac{27}{8}$	8

$A = \frac{1}{2} \times \frac{1}{2} \times (0 + 8 + 2(\frac{1}{8} + 1 + \frac{27}{8})) = \frac{17}{4}$

3 d)

$h = \frac{3-0}{6} = \frac{1}{2}$

x	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3
y	2	$\frac{17}{8}$	3	$\frac{43}{8}$	10	$\frac{141}{8}$	29

$A = \frac{1}{2} \times \frac{1}{2} \times (2 + 29 + 2(\frac{17}{8} + 3 + \frac{43}{8} + 10 + \frac{141}{8})) = \frac{429}{16}$

4 a)

$h = \frac{4-2}{4} = \frac{1}{2}$

x	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
y	14	20	27	35	44

$A = \frac{1}{2} \times \frac{1}{2} \times (14 + 44 + 2(20 + 27 + 35)) = \frac{111}{2}$

4 b)

$h = \frac{2-1}{4} = \frac{1}{4}$

x	1	$\frac{5}{4}$	$\frac{3}{2}$	$\frac{7}{4}$	2
y	4	$\frac{63}{16}$	$\frac{15}{4}$	$\frac{55}{16}$	3

$A = \frac{1}{2} \times \frac{1}{4} \times (4 + 3 + 2(\frac{63}{16} + \frac{15}{4} + \frac{55}{16})) = \frac{117}{32}$

4 c)

$h = \frac{4-1}{6} = \frac{1}{2}$

x	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	$\frac{7}{2}$	4
y	19	$\frac{71}{4}$	16	$\frac{55}{4}$	11	$\frac{31}{4}$	4

$A = \frac{1}{2} \times \frac{1}{2} \times (19 + 4 + 2(\frac{71}{4} + 16 + \frac{55}{4} + 11 + \frac{31}{4})) = \frac{311}{8}$

4 d)

$h = \frac{2--3}{5} = 1$

x	-3	-2	-1	0	1	2
y	4	3	4	7	12	19

$A = \frac{1}{2} \times 1 \times (4 + 19 + 2(3 + 4 + 7 + 12)) = \frac{75}{2}$

5 a)

$h = \frac{1--3}{4} = 1$

x	-3	-2	-1	0	1
y	1	10	11	10	13

$A = \frac{1}{2} \times 1 \times (1 + 13 + 2(10 + 11 + 10)) = 38$

5 b)

$h = \frac{2--2}{8} = \frac{1}{2}$

x	-2	$-\frac{3}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
y	5	$\frac{25}{4}$	7	$\frac{29}{4}$	7	$\frac{25}{4}$	5	$\frac{13}{4}$	1

$A = \frac{1}{2} \times \frac{1}{2} \times (5 + 1 + 2(\frac{25}{4} + 7 + \frac{29}{4} + 7 + \frac{25}{4} + 5 + \frac{13}{4})) = \frac{45}{2}$

5 c)

$h = \frac{2-0}{5} = \frac{2}{5}$

x	0	$\frac{2}{5}$	$\frac{4}{5}$	$\frac{6}{5}$	$\frac{8}{5}$	2
y	3	$\frac{73}{25}$	$\frac{67}{25}$	$\frac{57}{25}$	$\frac{43}{25}$	1

$A = \frac{1}{2} \times \frac{2}{5} \times (3 + 1 + 2(\frac{73}{25} + \frac{67}{25} + \frac{57}{25} + \frac{43}{25})) = \frac{116}{25}$

5 d)

$h = \frac{1--3}{5} = \frac{4}{5}$

x	-3	$-\frac{11}{5}$	$-\frac{7}{5}$	$-\frac{3}{5}$	$\frac{1}{5}$	1
y	11	$\frac{2081}{375}$	$\frac{1093}{375}$	$\frac{259}{125}$	$\frac{749}{375}$	$\frac{5}{3}$

$A = \frac{1}{2} \times \frac{4}{5} \times (11 + \frac{5}{3} + 2(\frac{2081}{375} + \frac{1093}{375} + \frac{259}{125} + \frac{749}{375})) = \frac{1132}{75}$

6 a)

$h = \frac{8-2}{4} = \frac{3}{2}$

x	2	$\frac{7}{2}$	5	$\frac{13}{2}$	8
y	24	$\frac{495}{8}$	84	$\frac{561}{8}$	0

$A = \frac{1}{2} \times \frac{3}{2} \times (24 + 0 + 2(\frac{495}{8} + 84 + \frac{561}{8})) = 342$

6 b)

$h = \frac{9-1}{4} = 2$

x	1	3	5	7	9
y	5	29	245	2189	19685

$A = \frac{1}{2} \times 2 \times (5 + 19685 + 2(29 + 245 + 2189)) = 24616$

6 c)

$h = \frac{-4--7}{6} = \frac{1}{2}$

x	-7	$-\frac{13}{2}$	-6	$-\frac{11}{2}$	-5	$-\frac{9}{2}$	-4
y	$\frac{8}{5}$	$\frac{14}{9}$	$\frac{3}{2}$	$\frac{10}{7}$	$\frac{4}{3}$	$\frac{6}{5}$	1

$A = \frac{1}{2} \times \frac{1}{2} \times (\frac{8}{5} + 1 + 2(\frac{14}{9} + \frac{3}{2} + \frac{10}{7} + \frac{4}{3} + \frac{6}{5})) = \frac{262}{63}$

6 d)

$h = \frac{7-5}{3} = \frac{2}{3}$

x	5	$\frac{17}{3}$	$\frac{19}{3}$	7
y	7	$\frac{23}{5}$	$\frac{25}{7}$	3

$A = \frac{1}{2} \times \frac{2}{3} \times (7 + 3 + 2(\frac{23}{5} + \frac{25}{7})) = \frac{922}{105}$

7)

$h = \frac{2-1}{4} = \frac{1}{4}$

x	1	$\frac{5}{4}$	$\frac{3}{2}$	$\frac{7}{4}$	2
y	0	0.1053	0.1731	0.2153	0.2393

$A = \frac{1}{2} \times \frac{1}{4} \times (0 + 0.2393\ldots + 2(0.1053\ldots + 0.1731\ldots + 0.2153\ldots)) = 0.153$ to 3 sf.

8 a)

$h = \frac{8-0}{4} = 2$

x	0	2	4	6	8
y	$\frac{k+0}{2}$	$\frac{k+4}{2}$	$\frac{k+16}{2}$	$\frac{k+36}{2}$	$\frac{k+64}{2}$

$$\begin{aligned} A &= \tfrac{1}{2} \times 2 \times (\tfrac{1}{2}k + \tfrac{1}{2}k + 32 + 2(\tfrac{1}{2}k + 2 + \tfrac{1}{2}k + 8 + \tfrac{1}{2}k + 18)) \\ &= k + 32 + 3k + 56 = 4k + 88 \end{aligned}$$

8 b)

$h = \frac{5-1}{4} = 1$

x	1	2	3	4	5
y	$k+1$	$\frac{k+2}{2}$	$\frac{k+3}{3}$	$\frac{k+4}{4}$	$\frac{k+5}{5}$

$$\begin{aligned} A &= \tfrac{1}{2} \times 1 \times ((k+1) + \tfrac{k+5}{5} + 2(\tfrac{k+2}{2} + \tfrac{k+3}{3} + \tfrac{k+4}{4})) \\ &= \tfrac{1}{2}(\tfrac{6k}{5} + 2 + 2(\tfrac{13k}{12} + 3)) = \tfrac{1}{2}(\tfrac{101k}{30} + 8) \\ &= \tfrac{101k}{60} + 4 = \tfrac{101k+240}{60} \end{aligned}$$

8 c)

$h = \frac{5-1}{4} = 1$

x	1	2	3	4	5
y	$\ln k$	$\ln 2k$	$\ln 3k$	$\ln 4k$	$\ln 5k$

$$\begin{aligned} A &= \tfrac{1}{2} \times 1 \times (\ln k + \ln 5k + 2(\ln 2k + \ln 3k + \ln 4k)) \\ &= \tfrac{1}{2}(\ln(5k^2) + 2\ln(24k^3)) = \tfrac{1}{2}\ln(5k^2) + \ln(24k^3) \\ &= \ln(\sqrt{5}k) + \ln(24k^3) = \ln(24\sqrt{5}k^4) \end{aligned}$$

8 d)

$h = \frac{5-1}{4} = 1$

x	1	2	3	4	5
y	$\ln \frac{1}{k}$	$\ln \frac{2}{k}$	$\ln \frac{3}{k}$	$\ln \frac{4}{k}$	$\ln \frac{5}{k}$

$$\begin{aligned} A &= \tfrac{1}{2} \times 1 \times (\ln \tfrac{1}{k} + \ln \tfrac{5}{k} + 2(\ln \tfrac{2}{k} + \ln \tfrac{3}{k} + \ln \tfrac{4}{k})) \\ &= \tfrac{1}{2}(\ln \tfrac{5}{k^2} + 2\ln \tfrac{24}{k^3}) = \tfrac{1}{2}\ln \tfrac{5}{k^2} + \ln \tfrac{24}{k^3} \\ &= \tfrac{1}{2}(\ln 5 - \ln k^2) + \ln 24 - \ln k^3 \\ &= \tfrac{1}{2}\ln 5 - \tfrac{1}{2}\ln k^2 + \ln 24 - 3\ln k \\ &= \tfrac{1}{2}\ln 5 - \ln k + \ln 24 - 3\ln k = \ln 24 + \tfrac{1}{2}\ln 5 - 4\ln k \end{aligned}$$

9)

$h = \frac{8-2}{3} = 2$

x	2	4	6	8
y	$\ln 7$	$\ln 5$	$\ln 3$	0

$A = \frac{1}{2} \times 2 \times (\ln 7 + 0 + 2(\ln 5 + \ln 3)) = \ln 1575 = 7.362$ to 3 dp. This will be an underestimate as the curve is concave between $x = 2$ and $x = 8$.

10)

$h = \frac{20-0}{4} = 5$

t	0	5	10	15	20
v	0	15	20	10	0

$s = \frac{1}{2} \times 5 \times (0 + 0 + 2(15 + 20 + 10)) = 225$ m.

11 a)

$h = \frac{1-0}{4} = \frac{1}{4}$

x	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
y	$\frac{1}{2}$	$\frac{4}{\sqrt{65}}$	$\frac{2}{\sqrt{17}}$	$\frac{4}{\sqrt{73}}$	$\frac{1}{\sqrt{5}}$

$A = \frac{1}{2} \times \frac{1}{4} \times (\frac{1}{2} + \frac{1}{\sqrt{5}} + 2(\frac{4}{\sqrt{65}} + \frac{2}{\sqrt{17}} + \frac{4}{\sqrt{73}})) = 0.481$ to 3 sf.

11 b)

Increase the number of strips.

12)

$h = \frac{1-0}{5} = \frac{1}{5}$

x	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
y	0	$\frac{\sqrt{70}}{42}$	$\frac{2\sqrt{665}}{133}$	$\frac{3\sqrt{190}}{76}$	$\frac{4\sqrt{105}}{63}$	$\frac{\sqrt{2}}{2}$

$A = \frac{1}{2} \times \frac{1}{5} \times (0 + \frac{\sqrt{2}}{2} + 2(\frac{\sqrt{70}}{42} + \frac{2\sqrt{665}}{133} + \frac{3\sqrt{190}}{76} + \frac{4\sqrt{105}}{63})) = 0.427$ to 3 sf.

13)

$s = \frac{1}{2} \times 3 \times (1 + 10.1 + 2(2.8 + 4.4 + 6.2 + 7.7 + 8.9)) = 106.65$ m.

14 a)

$h = \frac{4.5-2.5}{4} = \frac{1}{2}$ $A = \frac{1}{2} \times \frac{1}{2} \times (1.342 + 1.202 + 2(1.291 + 1.254 + 1.225)) = 2.521.$

14 b)

$$\int_{2.5}^{4.5} \sqrt{\tfrac{2+x}{2x}}\, dx = \tfrac{1}{\sqrt{2}} \int_{2.5}^{4.5} \sqrt{\tfrac{2+x}{x}}\, dx \approx \tfrac{1}{\sqrt{2}} \times 2.521 = 1.78 \text{ to 3 sf}$$

15 a)

$h = \frac{2-1}{4} = \frac{1}{4}$

x	1	$\frac{5}{4}$	$\frac{3}{2}$	$\frac{7}{4}$	2
y	e^3	$e^{75/16}$	$e^{27/4}$	$e^{147/16}$	e^{12}

$A = \frac{1}{2} \times \frac{1}{4} \times (e^3 + e^{12} + 2(e^{75/16} + e^{27/4} + e^{147/16})) = 23000$ to 3 sf.

15 b)

$$\int_1^2 e^{3x^2+1}\, dx = \int_1^2 e^{3x^2} e^1\, dx = e^1 \int_1^2 e^{3x^2}\, dx \approx e \times 23031.065\ldots = 62600 \text{ to 3 sf}$$

16)

$h = \frac{2-0}{5} = \frac{2}{5}$

$A = \frac{2}{5} \times (1490.4790 + 2796.1535 + 1786.4756 + 529.5354 + 49.6005 + 0.5000) = 2661.0976$

17 a)

$h = \frac{24-8}{4} = 4$ $A = \frac{1}{2} \times 4 \times (0.1787 + 0.8614 + 2(0.4307 + 0.6094 + 0.7481)) = 9.233.$

17 b)

$$\int_8^{24} \log_5 \tfrac{x^2}{36}\, dx = \int_8^{24} \log_5 (\tfrac{x}{6})^2\, dx = \int_8^{24} 2 \log_5 \tfrac{x}{6}\, dx = 2 \int_8^{24} \log_5 \tfrac{x}{6}\, dx \approx 2 \times 9.233 = 18.466$$

18 a)

$h = \frac{-0.5--2.5}{4} = \frac{1}{2}$

x	-2.5	-2	-1.5	-1	-0.5
y	2.1086	2.5713	2.7860	2.8845	2.9191

$A = \frac{1}{2} \times \frac{1}{2} \times (2.1086 + 2.9191 + 2(2.5713 + 2.7860 + 2.8845)) = 5.38$ to 3 sf.

18 b)

This will be an underestimate as the curve is concave between $x = -2.5$ and $x = -0.5$.

19)

$h = \frac{6-0}{5} = \frac{6}{5}$ $A = \frac{6}{5} \times (0.0000 + 0.4588 + 0.2872 + 0.1630 + 0.0903 + 0.0249) = 1.23$ to 3 sf.

20)

$h = \frac{\pi - 0}{5} = \frac{\pi}{5}$

x	0	$\frac{\pi}{5}$	$\frac{2\pi}{5}$	$\frac{3\pi}{5}$	$\frac{4\pi}{5}$	π
y	0	0.3454	0.9045	0.9045	0.3454	0

$A = \frac{1}{2} \times \frac{\pi}{5} \times (0 + 0 + 2(0.3454 + 0.9045 + 0.9045 + 0.3454)) = 1.57$ to 3 sf.

21 a)

$$\begin{aligned} A &= \tfrac{1}{2} \times 0.4 \times (p + 18 + 2(15 + 22 + p)) \\ &= \tfrac{1}{5}(p + 18 + 30 + 44 + 2p) = \tfrac{1}{5}(3p + 92) = \tfrac{3}{5}p + \tfrac{92}{5} \end{aligned}$$

21 b)

$\frac{3}{5}p + \frac{92}{5} = 25 \Rightarrow p = 11$

22 a)

$$\begin{aligned} A &= \tfrac{1}{2} \times (-4 - -20) \times (\ln(-20)^2 + \ln(-4)^2) \\ &= \tfrac{1}{2} \times 16 \times (2\ln 20 + 2\ln 4) \\ &= 8(2\ln(4 \times 5) + 2\ln 2^2) \\ &= 8(2(\ln 4 + \ln 5) + 4\ln 2) \\ &= 8(2(2\ln 2 + \ln 5) + 4\ln 2) \\ &= 8(4\ln 2 + 2\ln 5 + 4\ln 2) = 8(8\ln 2 + 2\ln 5) = 64\ln 2 + 16\ln 5 \end{aligned}$$

22 b)

$h = \frac{-4 - -20}{8} = \frac{16}{8} = 2$

$$\begin{aligned} A &= 2\ln(-4)^2 + 2\ln(-6)^2 + 2\ln(-8)^2 + 2\ln(-10)^2 \\ &\quad + 2\ln(-12)^2 + 2\ln(-14)^2 + 2\ln(-16)^2 + 2\ln(-18)^2 \\ &= 2(2\ln 4 + 2\ln 6 + 2\ln 8 + 2\ln 10 + 2\ln 12 + 2\ln 14 + 2\ln 16 + 2\ln 18) \\ &= 2(2\ln 2^2 + 2\ln(2 \times 3) + 2\ln 2^3 + 2\ln(2 \times 5) \\ &\quad + 2\ln(2^2 \times 3) + 2\ln(2 \times 7) + 2\ln 2^4 + 2\ln(2 \times 3^2)) \\ &= 2(4\ln 2 + 2(\ln 2 + \ln 3) + 6\ln 2 + 2(\ln 2 + \ln 5) \\ &\quad + 2(\ln 2^2 + \ln 3) + 2(\ln 2 + \ln 7) + 8\ln 2 + 2(\ln 2 + \ln 3^2)) \end{aligned}$$

$$
\begin{aligned}
&= 2(4\ln 2 + 2\ln 2 + 2\ln 3 + 6\ln 2 + 2\ln 2 + 2\ln 5 \\
&\quad + 2\ln 2^2 + 2\ln 3 + 2\ln 2 + 2\ln 7 + 8\ln 2 + 2\ln 2 + 2\ln 3^2) \\
&= 2(4\ln 2 + 2\ln 2 + 2\ln 3 + 6\ln 2 + 2\ln 2 + 2\ln 5 \\
&\quad + 4\ln 2 + 2\ln 3 + 2\ln 2 + 2\ln 7 + 8\ln 2 + 2\ln 2 + 4\ln 3) \\
&= 2(30\ln 2 + 8\ln 3 + 2\ln 5 + 2\ln 7) = 60\ln 2 + 16\ln 3 + 4\ln 5 + 4\ln 7
\end{aligned}
$$

23 a)

x	-5	-3	-1	3	5
y	$\frac{5}{6}$	$\frac{25}{14}$	$\frac{25}{6}$	$\frac{25}{14}$	$\frac{5}{6}$

$A_1 = 2(\frac{5}{6} + \frac{25}{14} + \frac{25}{6} + \frac{25}{14} + \frac{5}{6}) = \frac{395}{21} = 18.81$ to 4 sf.

23 b)

x	-3	-1	0	1	3
y	$\frac{25}{14}$	$\frac{25}{6}$	5	$\frac{25}{6}$	$\frac{25}{14}$

$A_2 = 2(\frac{25}{14} + \frac{25}{6} + 5 + \frac{25}{6} + \frac{25}{14}) = \frac{710}{21} = 33.81$ to 4 sf.

23 c)

$h = \frac{5--5}{5} = 2$

x	-5	-3	-1	1	3	5
y	$\frac{5}{6}$	$\frac{25}{14}$	$\frac{25}{6}$	$\frac{25}{6}$	$\frac{25}{14}$	$\frac{5}{6}$

$A_T = \frac{1}{2} \times 2 \times (\frac{5}{6} + \frac{5}{6} + 2(\frac{25}{14} + \frac{25}{6} + \frac{25}{6} + \frac{25}{14})) = \frac{535}{21} = 25.48$ to 4 sf.

23 d)

$$\int_{-5}^{5} \frac{25}{5+x^2}\,dx$$

$$x = \sqrt{5}\tan u \Rightarrow \frac{dx}{du} = \sqrt{5}\sec^2 u \Rightarrow dx = \sqrt{5}\sec^2 u\,du$$

Change limits:

$$x = 5 \Rightarrow 5 = \sqrt{5}\tan u \Rightarrow u = \arctan\tfrac{5}{\sqrt{5}} \Rightarrow u = \arctan\sqrt{5}$$

$$x = -5 \Rightarrow -5 = \sqrt{5}\tan u \Rightarrow u = \arctan(-\tfrac{5}{\sqrt{5}}) \Rightarrow u = \arctan(-\sqrt{5})$$

$$
\begin{aligned}
\int_{-5}^{5} \frac{25}{5+x^2}\,dx &= \int_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} \frac{25}{5+(\sqrt{5}\tan u)^2} \times \sqrt{5}\sec^2 u\,du \\
&= \int_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} \frac{25\sqrt{5}\sec^2 u}{5+5\tan^2 u}\,du
\end{aligned}
$$

$$
\begin{aligned}
&= \int_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} \frac{25\sqrt{5}\sec^2 u}{5(1+\tan^2 u)}\,du \\
&= \int_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} \frac{25\sqrt{5}\sec^2 u}{5\sec^2 u}\,du = \int_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} 5\sqrt{5}\,du \\
&= [5\sqrt{5}u]_{\arctan(-\sqrt{5})}^{\arctan\sqrt{5}} \\
&= (5\sqrt{5}\arctan\sqrt{5}) - (5\sqrt{5}\arctan(-\sqrt{5})) \\
&= (12.8603\ldots) - (-12.8603\ldots) = 25.72 \text{ to 4 sf}
\end{aligned}
$$